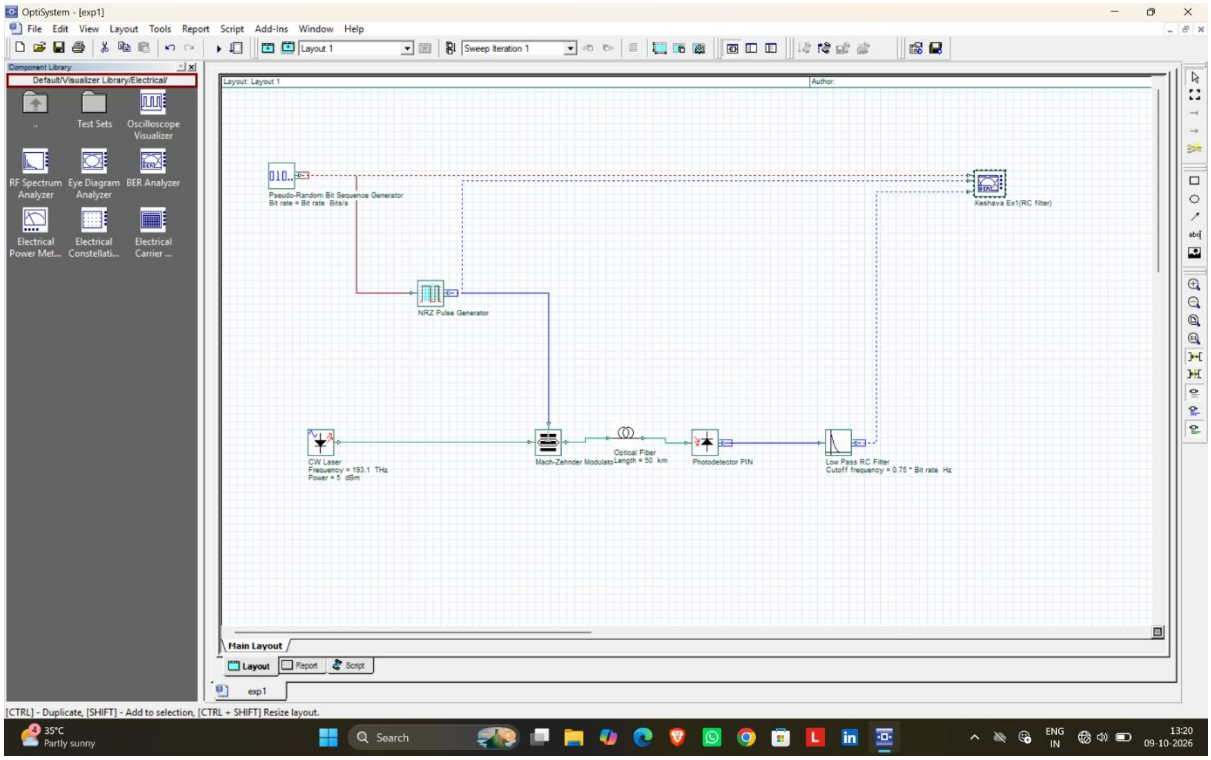
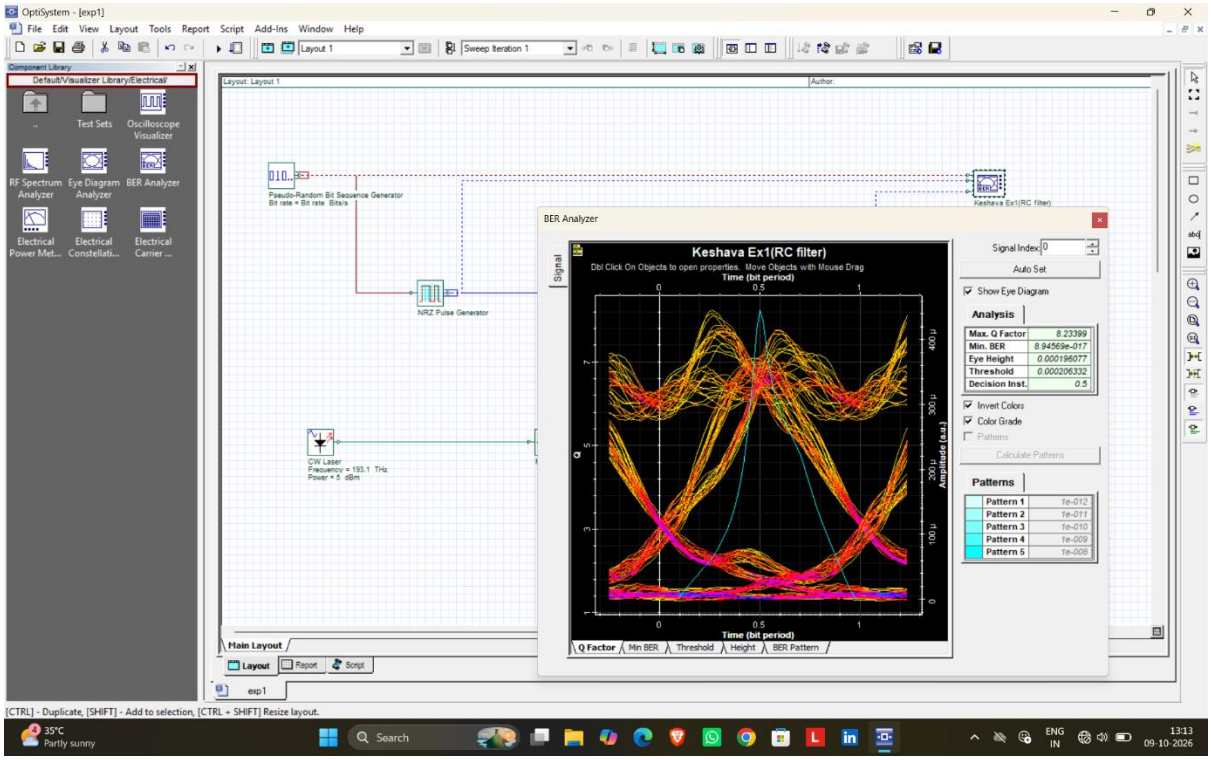
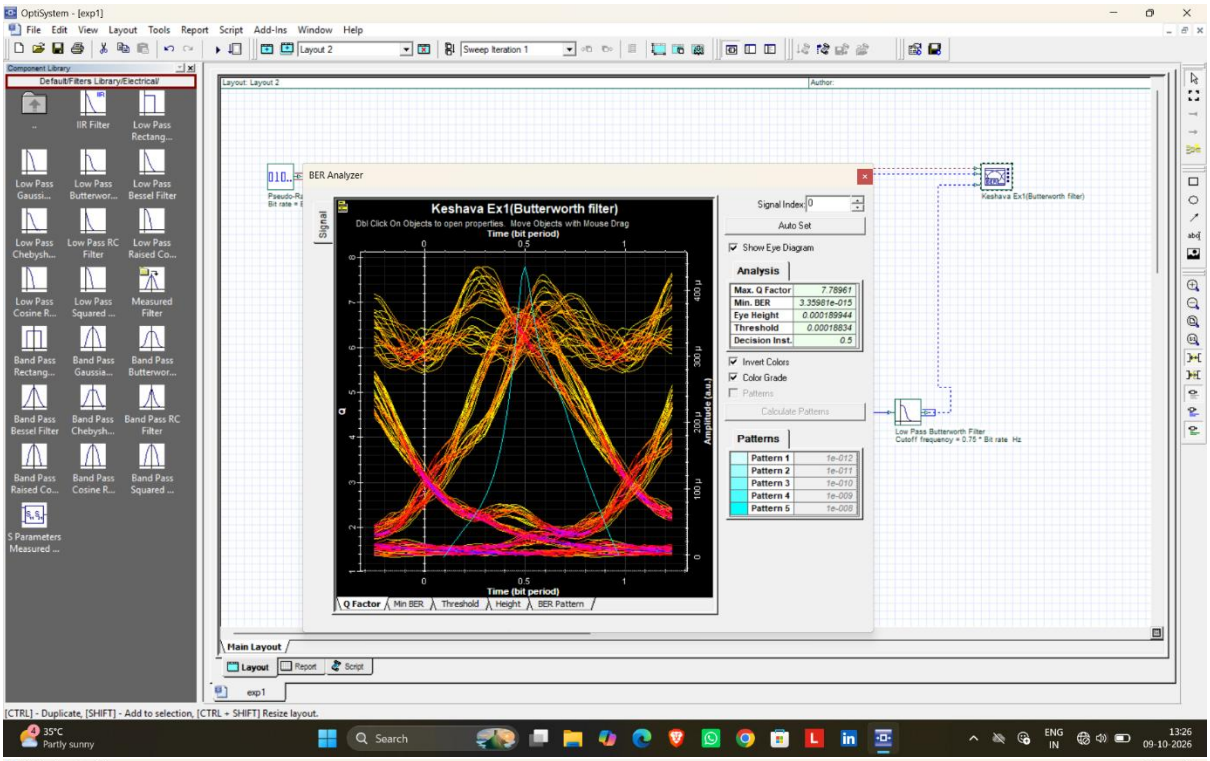
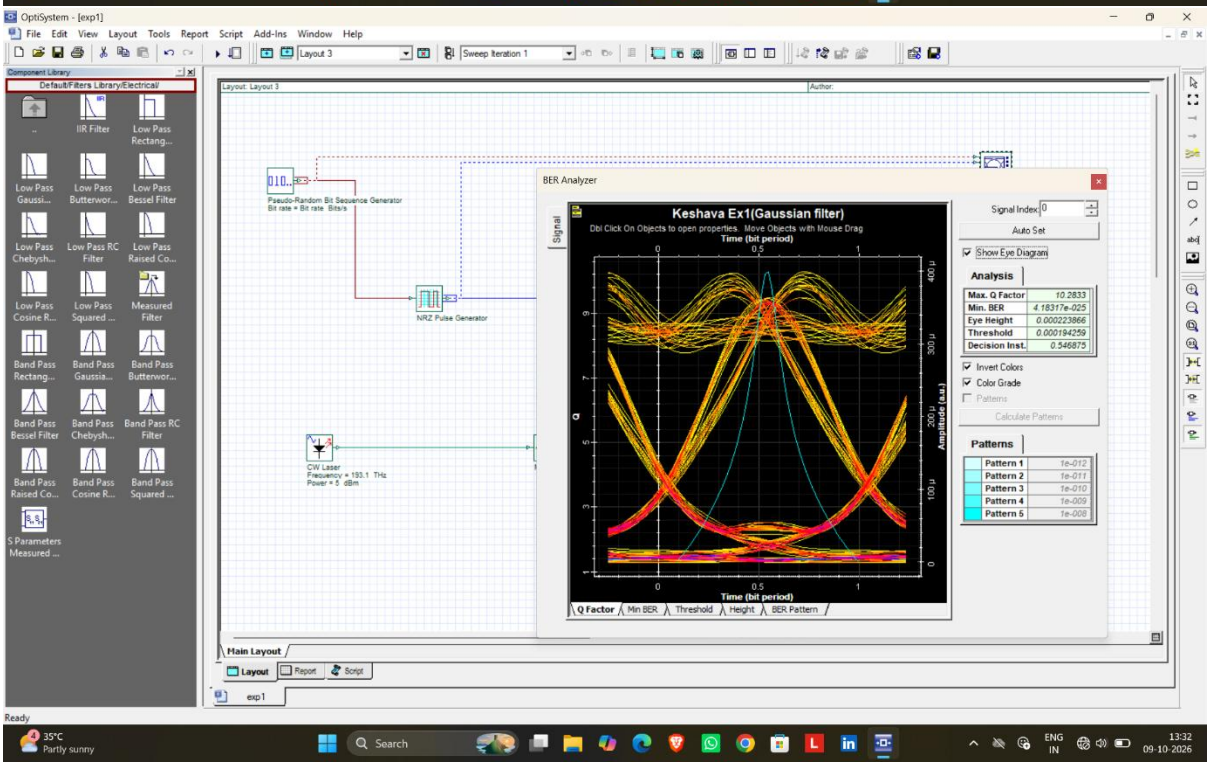
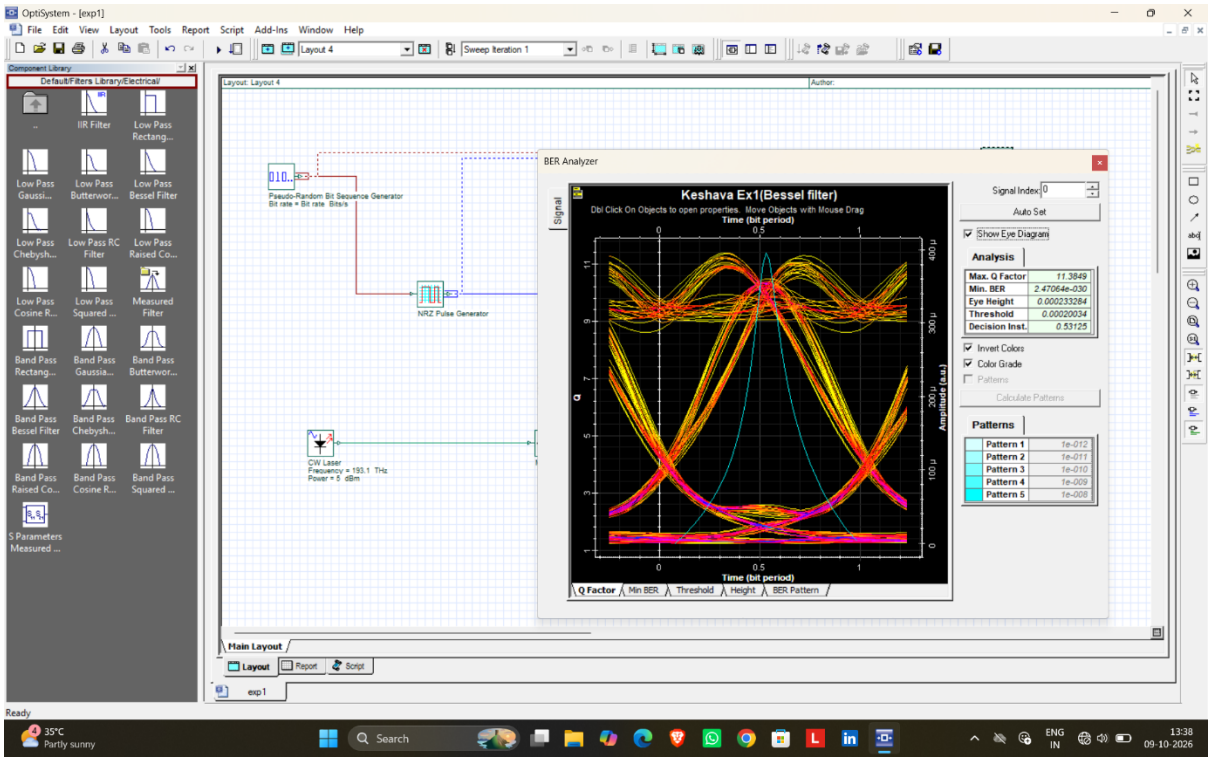
# EXP1: Fiber Optic Link Establishment and BER Analysis Using Optisystem




OptiSystem - [exp1]
File Edit View Layout Tools Report Script Add-Ins Window Help
Layout 2
Sweep Iteration 1
Default/Filters Library/Electrical/
BER Analyzer
Keshava Ex1(Butterworth filter)
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 7.78961
Min. BER 3.35981e-015
Eye Height 0.000189944
Threshold 0.00189834
Decision Inst. 0.5
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern
Keshava Ex1(Butterworth filter)
Low Pass Butterworth Filter
Main Layout
Layout
Report
Script
exp1
[CTRL] - Duplicate, [SHIFT] - Add to selection, [CTRL + SHIFT] Resize layout.
35°C
Partly sunny
Search
ENG
IN
13:26
09-10-2026


OptiSystem - [exp1]
File Edit View Layout Tools Report Script Add-Ins Window Help
Layout 3
Sweep Iteration 1
Default/Filters Library/Electrical/
Pseudo-Random Bit Sequence Generator
NRZ Pulse Generator
CW Laser
BER Analyzer
Keshava Ex1(Gaussian filter)
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 10.2833
Min. BER 4.18317e-025
Eye Height 0.000223866
Threshold 0.000194259
Decision Inst. 0.546875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern
Main Layout
Layout
Report
Script
exp1
Ready
35°C
Partly sunny
Search
ENG
IN
13:32
09-10-2026




**Output:**

| S.No | Filter name | Max Q factor | Min.BER | EYE Height | Thereshold |
|---|---|---|---|---|---|
| 1 | RC | 8.23399 | 8.94569e-017 | 0.000196077 | 0.000206332 |
| 2 | Butterworth | 7.78961 | 3.35981e-015 | 0.000189944 | 0.00018834 |
| 3 | Bessel | 11.3849 | 2.47064e-030 | 0.000233284 | 0.00020034 |
| 4 | Gaussian | 10.2833 | 4.18317e-025 | 0.000223866 | 0.000194259 |